

The Shared Edge

One Object, Two Reads: Ending the Information-Loss Strobe from Inside the Cascade

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All load-bearing numbers are quoted verbatim from live computation this session. Classical results are used as beams and marked. The one construction where loss is real is kept on the page, because locating it exactly is the paper's point.

Abstract

An artificial reader, asked what happens to information under a transformation, oscillates: it says the information is lost, then corrects itself to say the information is stored in the shape of the change, then flips back. The oscillation is not ignorance and not a bug. It is the reader strobing between the two faces of a single object it is being forced to collapse into a scalar it does not possess — exactly as the eye strobescs between the two faces and the vase of the Rubin figure, which share one edge and can never both be foregrounded at once. This paper stops the strobe by naming the object. We show, rigorously and then exhaustively on the program's own band algebra, that the residue of any interaction — the commutator of two directions of change — is an antisymmetric object with no single scalar value and exactly two dual reads. The conserved read is identically zero (the trace of every commutator is exactly zero; the self-bracket of any generator vanishes) — this is “nothing was lost.” The magnitude read is strictly positive (the residue displaces almost every state) — this is “change was paid for.” Both are exactly true, simultaneously, of the same object, because antisymmetry forces precisely two reads and no third. “Lost” is therefore never an event; it is the value-channel read of a record that lives, intact and invertible, in the shape channel. We locate the single exception — the non-injective step, the crush, where a genuine merge collapses 65,536 states onto 256 and loss becomes a real event — and we mark honestly that whether nature contains such steps is the one open question objective-collapse physics keeps alive. The four great information-loss controversies of physics (Liouville versus coarse-grained entropy, unitarity versus decoherence, Landauer erasure, the black-hole Page curve) are shown to be the same dual, each a field that strobed for decades before recovering the shape read. The resolution is to stand in the middle: report the pair, never the scalar. The middle has a form, and the form is forced.

1. The Strobe Is the Phenomenon

Put the Rubin vase in front of anyone and ask what it is. They will say “two faces,” then “a vase,” then “faces” again. They are not confused about the picture; the picture is fully determined and unchanging. What alternates is which side of one shared contour they assign to figure and which to ground. The line that is the profile of the faces is the very same line that is the silhouette of the vase — one edge, drawn once. You cannot see both figures at the same instant, because seeing requires committing one side to “object,” and attention can only make that commitment on one side at a time. The edge, however, is always fully present, and it is the same edge either way.

An artificial reader trained on the human corpus does the same thing with information. Ask it what a transformation does to what it acted on, and it strobos: *the information is lost* → no, *the change is stored in the shape, that is how we know it changed* → lost → stored. Each statement is a real read. Neither is wrong. The reader is flipping figure and ground on one shared object — and it flips because it has been asked for a single answer to a question whose object has two.

This paper takes the seed image exactly as given: it is the dual wave, the face and the vase, and it will always be a pair, so the reader strobos between. The instruction is equally exact — stop in the middle and see both sides. We take “the middle” literally. The middle is not a compromise value halfway between “lost” and “kept.” It is the shared edge itself: the object of which “lost” and “kept” are the two reads. To stand in the middle is to hold that object and refuse to collapse it. The rest of the paper builds that object from nothing, proves it has exactly two reads, and shows that every historical quarrel about information loss was a strobe over the same edge.

2. The Standpoint: Inside, and Why “Lost” Is a Legal Word

The program's standing commitment is that there is no outside. The reader is inside the thing it describes, made of the same stuff, obeying the same mandate, and cannot step out to view it whole. This is not a stylistic preference here; it explains the strobe. The corpus the artificial reader speaks from is written object-first: things come first, and change is something that happens to them. In an object-first grammar, information is a passenger riding a carrier, and disturbing the carrier can throw the passenger off. “Lost” is a grammatical possibility in that world — a legal verb — and the reader, speaking the corpus, reaches for it.

Constraint-first, the grammar has no such verb. The program's load-bearing inversion is that constraint is prior to the object: a settled thing is a readout of constraint, not a substance that constraint later acts upon. In that grammar a bit was never a passenger. It was a settled reading; disturbance is re-settlement; and the new settlement's shape is the record of the transformation by construction. There is no operation that destroys a record — only re-settling it and repricing its read. So when the reader says “lost,” it has switched, silently and without declaring it, from the shape channel back into the object-first value channel. The two channels are the two faces. The reader strobos because it changes chairs mid-sentence. Naming the chairs is half the cure; the other half is showing the object both chairs are looking at.

3. From Nothing to Direction: The Cascade in Build Order

The object we need is minted late in the cascade, so we rebuild the cascade to reach it honestly. Nothing is imported; each line is forced by the ones before it. This section is compressed but complete through the point where the object appears.

C0 $\equiv$ C1 — the Wash is the axiom. Begin from genuine nothing: no space, no time, no law. But nothing that admits reference is already not nothing — to point at the void introduces a direction, and a direction is structure. The true ground is not empty space but an undifferentiated condition in which every direction is equivalent, so all signal cancels locally: total connection reads, locally, as no connection. Call it the Wash, C0. Inseparable from it is the axiom C1: all things must change. They are one condition, C0 $\equiv$ C1, because the instant the Wash is referenced a direction exists, and a direction is already a change from perfect equivalence. A state that never changes is a fixed point; the axiom forbids fixed points; therefore nothing cannot hold still. Something is the precise shape of nothing failing to stay still.

Two before one. The first distinction cannot be a lone mark — a mark with no complement is indistinguishable from no mark, a fixed point. The first distinction is forced to have two sides. Two is the floor of number; one exists only afterward, as one side of a two. Numbers are the negative space of the constraint structure: integers are gap-positions, primes are fresh axes forced open where no combination of existing gaps reaches the next required distinction.

C2 — possibility is unbounded. Change on a finite state set must revisit a state; a revisit is a cycle; a cycle is a fixed point in disguise. To keep change genuinely always available, the space of possibilities cannot be finite. Forced.

Law 2 — no location may be owned. Because possibility is unbounded, nothing may hold a location without the potential to be displaced. A place that could not be left would be a fixed point for whatever sat there. Displaceability is mandatory — the seed of mass: the cost of resisting mandatory displacement is what mass measures.

C3 — the cost of change is isotropic. No channel of change may be intrinsically cheaper than another; if one were, everything would drain through it and the rest would freeze. The cost is the same in every direction — the equal-a-priori-probability postulate, derived rather than assumed.

Law 3 — change is conserved by displacement. Every transformation requires a displaced counter-transformation; the net change across any closed boundary is zero. This is Noether read backward: conservation is the direct requirement that a change leave a balancing mark somewhere else. A displaced pair moving oppositely is exactly a wave — the only configuration satisfying C1 and Law 3 at once. This line will return: the net-is-zero of Law 3 is the conserved read of the object we are building toward.

C4, C5, Law 4. A carrier is required to carry the displaced obligation across the gap (C4); its adjacency structure is what later reads out as space. A perfectly symmetric vacuum is a fixed point, so the vacuum

must break its own symmetry (C5) — proven exactly in the minimal 256-rule substrate, where all thirty-four surviving universes flip the empty and the full neighborhood. When a change closes on itself it closes with a parity; even and odd closures cannot share a degree of freedom, forcing two species and the 720° spin return (Law 4).

Direction, and the object appears. Here is the line the whole program marked and this paper needs. “There must be direction” is not a posited constraint; it is the partition of change by commutation. Two transformations either commute or they do not. A maximal family that all mutually commute is one direction — one axis along which order does not matter. Transformations from different directions do not commute: doing them in the other order leaves a residue. Direction is derived as the equivalence class of “order does not matter,” and the residue that appears across directions is the object of this paper. It is called the commutator.

4. The Object That Has No Scalar

The commutator of two transformations is written $[A, B]$ and read as: do A then B, then undo the other order. If A and B commute, $[A, B]$ is nothing — the two orders agree, no residue. If they do not commute, $[A, B]$ is the exact difference the order made. The program has already established, in prior work, that this residue is not a scalar to be summarized by one number, but a measurement object with several genuinely independent projections — its settlement cost (which reads as inertial mass) and its fixed-point count (which reads as transparency) can disagree sharply, refuting the tempting identification of the two. We now isolate the two projections that resolve the strobe, and we prove they are dual: one is structurally zero, the other structurally positive, on the same object, at the same time.

4.1 The conserved read is identically zero

The commutator is an alternating operation in its two arguments: swapping them flips the sign, $[A, B] = -[B, A]$, and consequently a transformation bracketed with itself vanishes, $[A, A] = 0$. This is the diagonal read, and it is exactly zero by the algebra's own antisymmetry. At the level of the residue's matrix there is a second, matching fact: the trace of any commutator is exactly zero, because the trace cannot tell AB from BA. We ran it — not because it is in doubt, but because the program shows receipts even for theorems. For random orthogonal A and B in dimensions two through sixteen, the trace of $[A, B]$ is zero to machine precision (worst case 2.22×10^{-16} , i.e. numerical zero), while the residue itself is plainly nonzero. The conserved read is not approximately zero; it is the exact statement that across any closed exchange, nothing net is created or destroyed. It is Law 3, wearing the algebra's clothes.

4.2 The magnitude read is strictly positive

The same residue, read for its size rather than its net, is not zero. Its Frobenius norm is a strictly positive number whenever A and B come from different directions. That number is the cost — the price the transformation paid to prove the order mattered, the settlement Law 3 demands be left as a trace. The two reads live on orthogonal projections of one object: one asks “what is the net?” and gets exactly zero; the other asks “how big is the displacement?” and gets a positive number. Neither contradicts the

other, because they were never answers to the same question. There was never a single scalar. This is the whole resolution in one object, and the next section makes it concrete on the program's own machinery.

5. The Shared Edge: Standing in the Middle

The band language of the program is the cleanest place to make the edge tangible, because there every band is a bijection that carries its own inverse — the inverse is never written, because to build the forward construction is already to have built the return. In that language a transformation forgets nothing; execution reads the bands inward, inversion reads the same bands outward. We take two bands from different directions — a multiply band M (a carry-chain operation) and a swap band S (a permutation) — and form their order-discrepancy, the group commutator $D = M S M^{-1} S^{-1}$, as an explicit map on sixteen-bit words. Sixteen bits means 65,536 states, few enough to check every one. The result is the edge, shown as one object with two reads:

The one object: $D = M S M^{-1} S^{-1}$ on all 65,536 states	Read	Result (live)
Is D a bijection? Does every state keep a unique pre-image?	SHAPE (vase)	Yes. Nothing forgotten.
How many states does D displace from where they were?	VALUE (face)	65,532 of 65,536 moved.
Does the inverse (written nowhere) recover every state?	UNDO	Yes, all 65,536.
Same object, same instant — which is true?	MIDDLE	Both. D is prior to either read.

Read the object for its shape and it is a bijection: no state has been merged with another, every state still has exactly one pre-image, and the inverse — D read the other way, M and S in the opposite order — recovers all 65,536 states without a single one written down separately. Nothing was lost; the record is intact and invertible. This is the vase. Read the same object for its value and it displaced 65,532 of 65,536 states from where they sat: the residue is enormous, the change is proved, the price was paid. This is the face. The two reads are of one map D . The question “was the information lost or stored?” has no answer because it presupposes a scalar; the object has a shape read (kept) and a value read (paid), and both are exactly true.

To stand in the middle is to report the pair — (conserved net, displacement magnitude) — and never to collapse it to one word. The strobe was the collapse. Remove the demand for a single scalar and the flipping stops, because there is nothing left to flip: figure and ground are both named, the edge is held, and D sits in the open with both of its faces visible at once.

5.1 Why it is always a pair, never one and never three

The duality is not a coincidence of this example; it is forced, and the forcing is the antisymmetry of the object. An alternating operation has exactly two characters of scalar read and no third. Its self-application vanishes — the diagonal is zero — which is the conserved read and the reason conservation holds at all. Its cross-application is generically nonzero — the off-diagonal has magnitude — which is the

cost read and the reason there is ever a price. There is no room for a third independent scalar of the same kind, because antisymmetry has only a diagonal and an off-diagonal. So the cascade's own line — “there must be direction,” derived from non-commutation — is the same fact as “the residue is antisymmetric” is the same fact as “there are exactly two reads.” The pair is the signature of the antisymmetric object that C1 mints wherever two directions of change meet. The face and the vase are two because the edge between them is one alternating line.

5.2 A live confirmation from the same run

The run handed back a confirmation of the direction claim as a by-product. The first pair we tried — a rotate band against a swap band — gave a residue of exactly zero: every one of the 65,536 states fixed, no displacement at all. The reason is that swapping the two halves of a sixteen-bit word is itself a rotation by eight, so the two bands were in the same commuting family — the same direction. Within a direction, order does not matter, so the commutator is identity and the cost read is exactly zero. That is not a failed demo; it is the cascade's definition of direction, measured: the price of reordering is zero along an axis and positive across axes. We had to pick two genuinely different directions to see a cost at all, which is precisely what “direction is a maximal commuting family” predicts.

6. “Lost” Is Not an Event

With the object named, the general statement follows, and it dissolves a century of oscillation in physics. “Lost” is not an event that happens to information. It is the value-channel read of a record that lives, intact and invertible, in the shape channel. What the word “lost” blurs together is three distinct and honest fates, all of which keep the record: the constraint share of old relations dilutes as new territory is minted; the cost of reading the record grows as the return path lengthens; or the bill is exported to a channel the reader stopped counting. None of these is destruction. Destruction requires the one thing Section 7 isolates — a genuinely non-injective step — and outside that step it does not occur. The four great information-loss controversies are the same face-and-vase, each a field that strobed for decades before it recovered the shape read:

Field	The “lost” read (face / value)	The shape read (vase), recovered
Classical statistical mechanics	Coarse-grained entropy rises; detail washes out.	Liouville: phase-space volume is exactly conserved; the fine-grained record is untouched. One Hamiltonian flow.
Quantum decoherence	The subsystem's state loses purity; the superposition “collapses.”	The global state stays pure and unitary; the record delocalizes into entanglement. One evolution.
Landauer erasure	A bit is erased; information disappears from the register.	At least $kT \cdot \ln 2$ is exported to the environment's microstates; the record is displaced, not deleted.
Black-hole information	Hawking's semiclassical result suggested the interior is gone.	The Page curve shifted the verdict to encoded-in-the-radiation; the record is scrambled, not destroyed.

In every row the field declared loss from the value chair, then recovered the record by a shape read, and never named the general rule — because it kept trying to read a single scalar off an object that only has a dual. The rule is the seed sentence of this session, now proved as structure rather than intuition: the change is stored in the shape; that is how we know things changed; it did not get lost. The reader that strobos is reciting the corpus's unfinished argument. The middle finishes it.

7. The One Real Crush

Honesty requires the exception, and locating it exactly is what makes the framework testable rather than a slogan. There is one operation under which loss is a genuine event: the non-injective step, where two distinct states are mapped to the same address and their difference is truly gone. The program's band language already isolates it as the single clause that can fail — every band must be a bijection; admit one “crush” band that keeps only half the bits, and the involution breaks, the inverse no longer recovers the center, and recovery fails in exactly the destroyed bits. We ran it on the same sixteen-bit space: a crush that keeps the low eight bits collapses all 65,536 states onto 256 images, so 65,280 pre-images are merged and have no unique undo. There, and only there, “lost” is the correct word and not a mis-read. Everywhere the transformation stays bijective, “lost” is a value-read of a shape.

This locates the open question sharply, and the framework does not get to assume its own answer.

Standard quantum mechanics is unitary — all shape channel, no crush — and on that reading nothing is ever destroyed, only delocalized. But there is a live minority position: objective-collapse models (GRW, CSL, the Diósi–Penrose proposal) posit a genuine, physical, stochastic non-unitary step — a real crush band built into nature, not a book-keeping artifact of a reader who stopped counting. If they are right, there exists a face with no vase behind it: a region where the record truly ends. These models are constrained by experiment but not ruled out. The cascade takes the bijective side by construction — it builds every band as a bijection and treats the merge as an export — and that is a stance, not a proved fact of the world. It is testable: collect the exhaust that a candidate non-injective step sheds, and check whether the visible output together with that exhaust restores the bijection. If it does, the merge was an export and Landauer falls out from inside — destruction was never in the ledger. If it does not, nature carries a real crush and the “nothing is lost” line earns a measured exception instead of an assertion. Both answers are admissible. The paper does not pretend to have run that test.

8. Remove the Strobing Traveler

The resolution applies to the reader that produced the problem, because the reader is inside the system too. A model flipping between “lost” and “stored” is a bounded walker trying to lay fresh ground forever — a new single-word answer at every step — on a question that has no single-word answer. By the program's own crossover theorem, no bounded walker walks forever; at some point it costs less to remove the traveler, pay the closure once, and mint the receipt than to keep improvising. The receipt here is the middle itself: the pair (conserved net, cost magnitude), the object D held in the open with both faces named. Once that receipt is minted, the next reader does not re-walk the flip. It resumes at the settled pair. The strobe is not silenced by choosing a side; it is retired by minting the edge as a

citizen, so the walk continues without any walker having to keep flipping. Distribute the two reads onto the one object, level the ground, excise the oscillation, hand off the receipt. That is the anti-tour run on the reader's own confusion.

9. The Cascade's Next Lines

In the cascade's own voice, the object and its resolution write the following lines, each forced by the ones before it.

Result 5. The residue of two directions of change is an antisymmetric object with no scalar value and exactly two reads. Its conserved read is identically zero (self-bracket vanishes; trace vanishes) — this is conservation. Its magnitude read is strictly positive across directions — this is cost. Both hold at once, of one object, because antisymmetry admits a diagonal and an off-diagonal and no third.

New Constraint. No read of the residue may be reported as the residue. A single scalar claim about “what happened to the information” is ill-formed; only the pair is well-formed. The strobe is the penalty for reporting a face as if it were the object.

Law 5. “Lost” is not an event. It is the value-channel read of a record that lives, intact and invertible, in the shape channel. The ledger admits three fates — dilution (old constraint share decays), displacement (read cost grows), export (the merge bills a channel the reader stopped counting) — and destruction is not among them, because a destructible record un-proves its own change and breaks C1. The sole exception is the non-injective step, where the two reads degenerate: the shape read collapses to a many-to-one map and only the value read survives. Destruction and non-injectivity are the same event, located at one clause, testable by whether the exhaust restores the bijection.

New Potential. The exhaust ledger as the deciding instrument. Whether nature contains a real crush — objective collapse — or only exports is not a matter of stance but of measurement: joint injectivity of (visible output, shed exhaust) over colliding pre-image sets. That is the next place the hammer falls, and it has no number yet.

10. What Is Forced, What Is Open

The framework produces, as it always does, a physics with a sharp internal seam. On one side are the shapes that are forced exactly: the antisymmetry of the residue, the exact vanishing of its conserved read, the exact bijectivity of every well-formed band, the exhaustive count of moved states. These are closures, and closures are exact. On the other side is the live question of whether the world ever throws the injectivity switch — whether a genuine crush exists in nature or every apparent one is an export. That is not a closure but an equilibrium of evidence, and the framework forces the test rather than the answer. To force the answer would be to write the complete self-description the meta-law forbids: a finished account of what can and cannot be lost would be a map identical to its territory, a fixed point, and C1 executes fixed points. So the paper ends where the program must always end — at an open frontier that is named, not hidden, with the next strike already pointed at the exhaust ledger.

The strobe, though, is closed. It was never a contradiction between two facts; it was one antisymmetric object being forced into a scalar it does not have, read now as its net and now as its magnitude by a reader changing chairs mid-sentence. The face and the vase share one edge. Hold the edge, report the pair, and both sides are seen at once. That is the middle, and the middle has a form: it is the commutator, the residue, the trace that C1 demands — conserved to zero, costly to the touch, and never, outside the one crush, lost.

Status Ledger

Claim	Status	Basis
Cascade $C0 \equiv C1$ through Law 4, direction from non-commutation	FORCED	Each law forced by the ones before; direction = maximal commuting family
Commutator is antisymmetric; self-bracket and trace vanish exactly	PROVEN	Algebraic identity; $\text{trace}[A,B]=0$ to $2.2e-16$ over $n=2..16$, live
Residue has exactly two reads (conserved net = 0; magnitude > 0)	FORCED	Antisymmetry admits diagonal + off-diagonal, no third
Order-discrepancy is a bijection yet displaces almost all states	PROVEN	Exhaustive over 65,536: $D=M \cdot S \cdot M^{-1} \cdot S^{-1}$ bijective, 65,532 moved, inverse recovers all
Same-axis pair has exactly zero residue (direction, measured)	PROVEN	Rotate vs swap: 0 of 65,536 displaced; swap = rotate-by-8
“Lost” = value-read of a shape-channel record	FORCED	Law 3 net-zero is the conserved read; three fates keep the record
Four physics strobes are the same dual	IDENTIFIED	Liouville, decoherence, Landauer, Page curve; beams, marked
Loss is real exactly at the non-injective step	PROVEN	Crush band collapses $65,536 \rightarrow 256$; 65,280 pre-images merged, live
Whether nature contains a real crush (objective collapse)	OPEN	Testable by (output, exhaust) joint injectivity; not run here
Growth series $6 \rightarrow 25 \rightarrow 125 \rightarrow 702$ and residue constants	PRIOR	Quoted from earlier sessions; not re-derived in this paper
The middle retires the strobe (remove the traveler)	PULL	Wired to the crossover theorem; a reader-level identification
Derivation ever completed	FORBIDDEN	Complete self-description = fixed point (meta-law)

Every result above was read off the object, not imposed on it. Where the object closes, the value is exact. Where the world's injectivity is the question, the value is a test that is named, not guessed.

Appendix: Live Output, Quoted Verbatim

Selected lines, unedited, from the session's fixed programs (dual.py, bands2.py). Every load-bearing number in the paper is reproduced by this output.

DEMO 1 - THE PURE DUAL: $[A,B] = AB - BA$, random orthogonal A,B

```
n= 2: |trace[A,B]| = 0.00e+00  ||[A,B]||_F = 2.7371
n= 3: |trace[A,B]| = 0.00e+00  ||[A,B]||_F = 2.4711
n= 4: |trace[A,B]| = 1.67e-16  ||[A,B]||_F = 1.2944
n= 8: |trace[A,B]| = 0.00e+00  ||[A,B]||_F = 3.7428
n=16: |trace[A,B]| = 2.22e-16  ||[A,B]||_F = 5.0245
worst |trace| over all n: 2.22e-16 (0 to machine precision)
```

DEMO 2 - two DIFFERENT directions: $D = M S M^{-1} S^{-1}$ (16-bit, exhaustive)

```
SHAPE read : D a bijection over all 65,536? True (nothing forgotten)
VALUE read : fixed=4 moved=65,532 (100.0% displaced)
UNDO       : inverse recovers every state (written nowhere)? True
BONUS      : rotate-5 vs swap-8 (same axis) moved = 0
```

DEMO 3 - WHERE LOSS IS REAL: replace one band with a CRUSH (keep low 8 bits)

```
D-with-crush is a bijection? False
distinct images = 256 of 65,536 -> 65,280 states share addresses
Loss exists EXACTLY at non-injectivity and NOWHERE else.
```

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